

Dhanush Munagala

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SUMMARY

Data Analyst completing M.S. Management Information Systems at the University of Houston (4.0 GPA, Dean's Award, May 2026) with production experience in SQL, Python, R, Power BI, Snowflake, and Airflow. Built clinical machine learning models achieving 85% AUC on real patient data, delivered market intelligence and data visualization analytics quantifying a \$760B industry across 132 U.S. facilities, and engineered automated data pipelines with a 30% efficiency gain. Available for full-time – June 2026.

Technical Skills

Languages & Querying: SQL, Oracle SQL, MySQL, Python (pandas, NumPy, scikit-learn), R(dplyr, ggplot2), Big Data, RDBMS

Analytics & Modeling: Statistical Analysis, Predictive Modeling, Machine Learning, Cohort Analysis, PCA, A/B Testing, Forecasting, EDA

BI & Visualization: Power BI (DAX, Star Schema, Custom KPIs), Tableau (LOD Expressions), Advanced MS Excel (Macros, Pivot Tables, VLOOKUP/XLOOKUP, Power Query), Data Visualization, Business Intelligence, Data storytelling, KPI Development, Dashboard Development

Data Engineering: ETL/ELT Pipeline Design, Data Modeling, Data Wrangling, Data Validation, Data Quality, Database Management

Tools & Platform: Snowflake, Apache Airflow, Alteryx, GitHub, Agile/Scrum, Smartsheet, JMP, Stakeholder Communication, MS Office Suite

PROFESSIONAL EXPERIENCE

UH Bauer Gutierrez Energy Management Institute × Cognitiv AI (FD AI & Tech Solutions), Houston, TX **Feb 2026 – May 2026**

Data Analytics & Market Intelligence Consultant | Faculty Supervised

- Engineered a market sizing and segmentation framework across 132 U.S. refineries using IBIS World, Barnes Reports, EIA, and Oxford Economics data; performed ad hoc analysis and developed NCI-based data model delivering TAM/SAM/SOM quantification for a client entering a \$760B market, improving beachhead targeting precision by 18%.
- Designed and deployed a **Power BI safety analytics dashboard** on a 5-year OSHA dataset (2020–2024); automated **injury trend detection, severity classification, and YoY comparisons** across 10 refinery sites, converting raw incident records into cross-functional decision-ready intelligence for client and faculty stakeholders.

C. T. Bauer College of Business, UNIVERSITY OF HOUSTON, Houston, TX

Jan 2026 – May 2026

Teaching Assistant – IT Project Management & Logistics Management

- Delivered structured instruction and performance feedback to **270+ students across 6 sections** in demand forecasting, inventory optimization, supplier performance analysis, and supply chain network design; facilitated **Harvard Business School 3M Canada Healthcare Supply Chain case study** and **Harvard's Project Management Simulation V3**.

C. T. Bauer College of Business, UNIVERSITY OF HOUSTON, Houston, TX

Jul 2025 – Jan 2026

Teaching Assistant – Business Applications of Database Management (Oracle SQL)

- Mentored **190+ students** in advanced **Oracle SQL – Relational schema design, 3NF normalization, complex joins, correlated subqueries, window functions, and query optimization**; reduced project error rates through structured debugging sessions.

R G SOFTWARE, Hyderabad, India

Jul 2023 – Jul 2024

Associate Data Analyst

- Worked within a **production data engineering team** operating **Python, SQL, Apache Airflow, and Snowflake**; contributed to extraction, transformation, and loading (ETL) of **large-scale, multi-source operational datasets** from **Oracle source systems**; automated recurring data preparation workflows using **Alteryx Designer**, reducing manual preparation time by 30%.
- Gathered **business reporting requirements** from 3 client accounts and translated them into **scalable analytical deliverables**; maintained **data integrity** by building automated **Python validation checks** that identified and flagged data quality issues, improving data accuracy by approximately 15–20%; developed **Tableau dashboards** with LOD expressions tracking cost variance, utilization, and delivery SLAs for 20+ stakeholders; produced **15+ monthly analytical reports**, reducing reporting overhead by 30% and ensuring consistent on-time delivery across all client accounts.

EDUCATION

UNIVERSITY OF HOUSTON, C. T. BAUER COLLEGE OF BUSINESS, Houston, TX

Aug 2024 – May 2026

Master of Science in Management Information Systems, GPA: 4.0/4.0 | **Dean's Award**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, Hyderabad, India

Jul 2019 – Jul 2023

Bachelor of Technology in Electronics and Communication Engineering, GPA: 3.5/4.0 | **Gold Medalist** (#1 Academically)

PROJECTS

Clinical Machine Learning Model: Diabetes Progression & Comorbidity Prediction [Python, JMP Pro, R]

Sep 2024 – Dec 2024

- Developed a clinical machine learning pipeline on a real 442-patient dataset, **feature engineering, PCA, logistic regression, and decision trees**, achieving 85% AUC (diabetes), 0.72 AUC (cardiovascular risk), and 0.71 AUC (kidney disease); applied **causal inference analysis (chi-square, ANOVA)** to identify statistically significant intervention levers across 3 comorbidity pathways.

Production RDBMS Design & SQL Analytics [Oracle SQL, Schema Design, Data Modeling]

Aug 2025 – Dec 2025

- Designed a **production Oracle SQL RDBMS** with 3NF normalization, referential integrity, FK constraints, and domain-level validation across 8+ entities; engineered **40+ analytical SQL queries** covering complex joins, correlated subqueries, window functions, and hierarchical queries to generate operational performance metrics.

Leadership

President, Bauer Entrepreneurship Club (Aug 2025 – May 2026) | **Graduate Ambassador**, Bauer College of Business (Aug 2024 – Dec 2025)